

Annotated Bibliography

Wasser, D. (2026, June 2). *A new measure of scientific disruption provides an innovative way to identify and credit research breakthroughs*—University of Virginia School of Data Science.

<https://datascience.virginia.edu/news/discovery-didnt-count>

New Measurement of Scientific Disruption | Identify and credit research breakthroughs

In the past, researchers measured a paper's impact by **how many times it was cited**. In 2017, the *Consolidation-Disruption (CD) Index* was created to address this issue. The goal was to identify when a paper **was disruptive** (the contribution was so distinct it counted as a breakthrough and eclipsed those previous)

But the *CD Index* has a blind spot. If 2 research teams discover something simultaneously and one cites the other, the index will rank one paper above the other, giving it breakthrough status and the other a lower rank. Or, in some cases, both at the bottom.

- If later citers mostly cite the focal paper without also citing its predecessors → the paper looks "disruptive" (it made the earlier work obsolete) → score near +1.
- If later citers cite the focal paper and its predecessors together → the paper looks "consolidating" (it's just building on/alongside prior work, not replacing it) → score near -1.

The *CD Index* score doesn't ask "was this a breakthrough?" — it asks "did later papers stop citing the old stuff once they started citing this one?" So both papers look the same to the index: **Later work still cites the old stuff alongside this paper = not disruptive = low score**. If both are released simultaneously, they will likely be cited in the future and will be counted as consolidating rather than disruptive.

Thus, they published a different citation ranking system, **EDM (Embedding Disruptiveness Measure)**.

EDM uses techniques from neural language models to give each paper two representations, one for what came before it and one for what came after.

"Nobel-level work (groundbreaking) tends to score significantly higher under EDM; it's stable against small citation changes, and it can identify simultaneous discoveries the older method couldn't."

The reason for this is to hopefully apply it to large-scale studies, including "research on how breakthroughs happen," to know what:

- Kinds of teams produce them
- Funding relating to disruptiveness
- More robust, so more reliable
- Crediting simultaneous discoveries

Kim, M., Kojaku, S., & Ahn, Y.-Y. (2026). **Uncovering simultaneous breakthroughs with a robust measure of disruptiveness**. *Science Advances*, 12(14)

[eadx3420. https://doi.org/10.1126/sciadv.adx3420](https://doi.org/10.1126/sciadv.adx3420)

The Core Concept: The "Two Vector" Approach

Instead of plotting a scientific paper as a single dot in space, EDM gives every single paper two distinct coordinates (vectors):

The Past Vector: Represents a paper's intellectual foundation—the previous papers it cites (its antecedents).

The Future Vector: Represents a paper's legacy—the subsequent, newer papers that cite it (its descendants).

How it's Done

1. Random Walks: AI goes from paper to paper, following citations, creating long chains.
2. Skip-gram Model:

$$\mathcal{J} = \sum_{s \in V} \sum_{r=1}^R \sum_{t=1}^{T-c} \sum_{w=1}^c \log \Pr \left[v_{t+w}^{(r,s)} \mid v_t^{(r,s)} \right]$$

Where V represents the set of all papers, and $s \in V$ denotes a paper from which a random walk is initiated. For each paper s , R random walks are performed, where r indexes the random walks, and t is the position within each random walk of length T . The context window

size is denoted by c , and w represents the citation step within the window. The term $v_t^{(r,s)}$

refers to the node at position t in the r th random walk starting from paper s , whereas $v_{t+w}^{(r,s)}$

is the antecedent paper of $v_t^{(r,s)}$ within w citation steps.

The algorithm tries to predict "what comes next" and "what came before" in those citation chains. As it trains, it shifts the vectors in the high-dimensional space.

3. Measuring Disruptiveness: Once all the papers have settled into their positions in the space, the algorithm measures the cosine distance between a single paper's own past vector and future vector.

Developing (Consolidating) Papers: If a paper simply builds on a continuous stream of thought, the newer papers that cite it will also cite its older foundational papers. Because the past and future rely on the same network, the paper's past and future vectors are pulled close together in space.

Disruptive Papers: If a paper introduces a massive paradigm shift, the newer papers that cite it will focus on the new idea and stop citing the old foundational papers. Because the future knowledge eclipses the past, the paper's past and future vectors drift far apart. The larger the distance between these two points, the more "disruptive" the paper is.

Kim, M., Kojaku, S., & Ahn, Y.-Y. (2025). *Uncovering simultaneous breakthroughs with a robust measure of disruptiveness*. arXiv.

<https://arxiv.org/abs/2502.16845>

Section Overviews

Abstract

States that scientific and technological breakthroughs are graded on a scale of 'disruptiveness.' Current citation methods (Disruption Index) do not accurately attribute these disruptive breakthroughs (due to simultaneous discoveries, Nobel Prize-winning papers, etc.), and the goal is to quantify this with a

more specific, in-depth metric. Thus, this paper introduces a new model of Embedding Disruptive Measure.

1. Introduction

Current *Disruption Index* practices use a ‘Bow-tie Structure’ that severs the stream of marginal innovations to create a new stream when one paper deems the previous ones obsolete. Widely adopted. However, it's a *discrete* measure that relies on the local citation network.

1. Small changes to the topology change a lot
2. Only ranks papers into 3 categories
3. Only examines papers in the local vicinity of the focal paper
4. Cannot tell the difference between simultaneous events

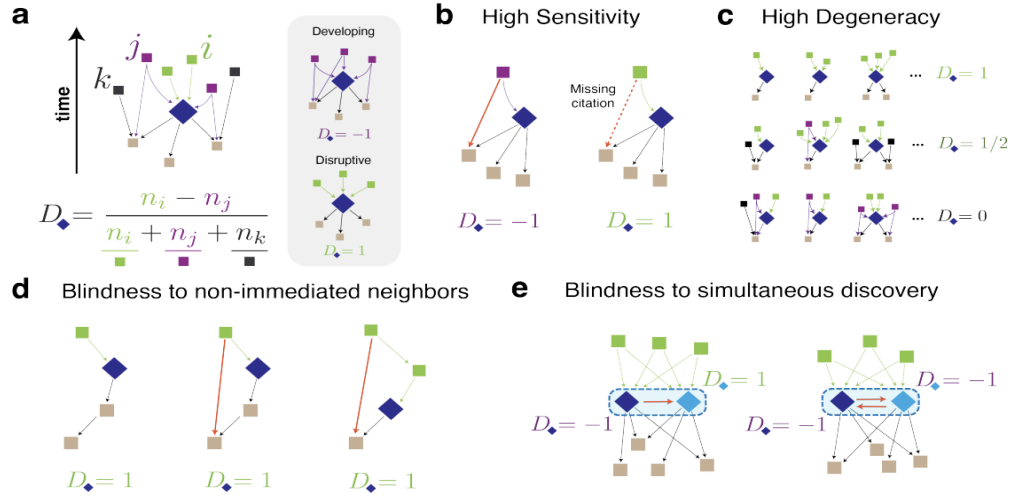
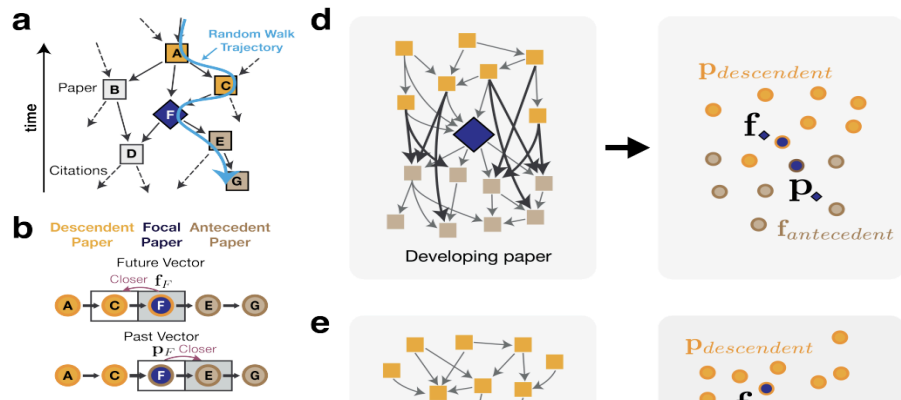


Figure 1: **The disruption index has critical limitations due to its discreteness and locality.** (a) The disruption index quantifies the degree to which the descendent works rely solely on a focal work and are free from antecedent works. The disruption index of a focal paper reaches its minimum value $D = -1$ when all future works citing the focal work also cite the prior works referenced by the focal work. On the other hand, the focal work is maximally disruptive ($D = 1$) when all future works cite the focal work while not citing any past works referenced by the focal work. (b) The disruption index is extremely sensitive; even a single missing citation can cause it to shift dramatically from -1 to 1. (c) D exhibits high degeneracy, having the same values for the different citation topology structures. (d) The index only captures the local structure formed by directly connected papers, neglecting any structure beyond the immediate vicinity. (e) In cases where two papers jointly or simultaneously create disruption and receive equal recognition from descendants, even a single citation can turn them minimally disruptive from maximally disruptive.

2. Results

The results section outlines how directional random walks and skip-gram optimization construct past and future vectors across the entire citation network to measure disruptiveness.



2. 1. Denegeracy and Locality

The measure is denoted as Δ and is continuously defined by past citations (rather than losing them); thus, it is not degenerate. Thus, with the same paper used in D (the discrete previous citation technique), the new measure can go beyond 2-3 and further papers.

2. 2. Identification of Disruptive Papers

This subsection compares EDM with the disruption index in identifying landmark research. The study used American Physical Society (APS)- selected Nobel Prize papers.

2. 2. 1. Nobel Prize and Milestone Paper

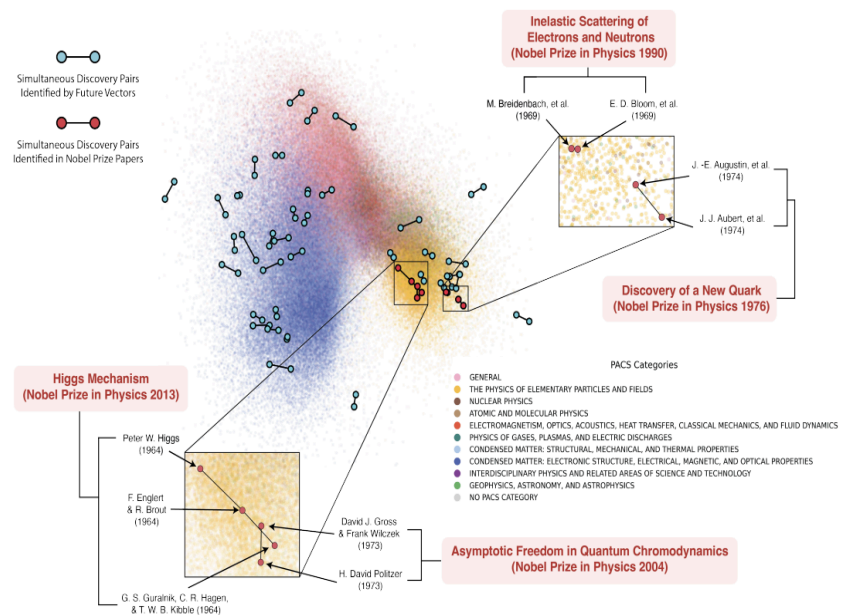
The value of Δ is closely linked to the chance that a paper will be designated as a milestone paper. For the Δ percentile, the odds ratio was 1.23 ($p < 0.001$; 95% CI: 1.14–1.34), meaning a 10% increase in the Δ percentile is associated with about a 23% increase in the odds of the paper being classified as a milestone paper. For citations, the relationship was even stronger, with an odds ratio of 6.14 ($p < 0.001$; 95% CI: 5.51–6.87), meaning that when a paper's citations roughly double, its odds of being designated as a milestone paper increase by about 6.14 times.

2. 2. 2. Simultaneous Disruption

Understanding why the D measure has lower-than-expected scores for milestone papers, they examined the top 10 different scores of D and Δ to see why. They found that when two research teams independently made the same breakthrough and their papers cited each other, even one citation could cause D to drop dramatically.

2. 2. 3 Identification of Simultaneous Discoveries

By analyzing nearest neighbors among thousands of papers published in the same year, the method successfully retrieves and annotates dozens of independent and collective simultaneous discovery pairs. In quotes from the text: “Figure 4: Future vectors reveal the simultaneous discoveries. The PCA plot of the future vectors of 332,518 APS papers. Each paper’s future vector is color-coded according to its PACS category. Simultaneous discovery papers identified in the Nobel Prize-winning papers dataset are represented by large red dots, with their closest neighbors connected by black edges. Two papers involved in the discovery of reverse transcriptase are excluded as they are not included in the APS dataset. For the Higgs mechanism’s simultaneous discovery, we included an additional paper by Guralnik et al. [39] (see Supplementary Section 4 for details). Notably, the future vectors of these simultaneous discovery papers are the closest pairs to each other. Using this feature, we identified 18,417 potential simultaneous discovery papers, among which we annotated 80 papers with more than 300 citations. Of the 80 annotated papers, 64 were identified as simultaneous discovery papers: 34 papers are independent simultaneous discovery papers, where the authors explicitly reference their



simultaneous discovery counterparts within the text, while another 30 papers represent collective simultaneous discoveries, where the same authors shared their findings across multiple publications. The 34 independent simultaneous discovery papers are colored with bright teal in the figure. The black edge again shows that the future vectors of these simultaneous paper pairs are closest to each other. Principal components 2 and 3 are shown, while principal component 1 primarily captures temporal information.”

Funk and Owen-Smith, *A Dynamic Network Measure of Technological Change*.

Jay Alammar, *The Illustrated Word2vec*.